DATABASE MANAGEMENT SYSTEM

Chapter 2: Data Models

CT 652 | BE BCT | Year III / Part II | TU IOE | Hours: 7 | Marks Weightage: 12

Syllabus Overview

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2.1 Data Models — Logical, Physical and Conceptual

A data model is a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints. It provides a way to describe the design of a database at the physical, logical, and view levels.

Types of Data Models

Type	Description
Conceptual (High-level) Data Model	Describes data in terms of real-world entities and relationships. Close to how users perceive data. Example: E-R Model.
Logical (Representational) Data Model	Describes data in terms of data structures such as tables, rows, columns. Example: Relational Model, Hierarchical, Network.
Physical (Low-level) Data Model	Describes how data is stored on disk — files, indexes, access methods. Closest to actual storage.
View (External) Data Model	Describes a subset of the database relevant to a particular user group. Hides the rest.

Exam Tip: "What are data models? Explain various types of data models." [1+3] — commonly asked.

2.2 E-R Model (Entity-Relationship Model)

The Entity-Relationship (E-R) Model was proposed by Peter Chen in 1976. It is a high-level conceptual data model used to represent the structure of a database in terms of entities, attributes, and relationships. It serves as the basis for designing a relational database.

Components of the E-R Model

- Entity: A real-world object or thing that is distinguishable from other objects.
- Attribute: Properties or characteristics that describe an entity.
- Relationship: An association between two or more entities.
- Key: One or more attributes whose values uniquely identify an entity.

Exam Tip: "Explain the various elements of the relational data model along with an example." [3+1] — The E-R model components are among the most tested topics. Worth 12 marks in the syllabus.

2.3 Entities and Entity Sets

Entity

Entity: An entity is a real-world object or thing that is distinguishable from other objects. An entity can be a person, place, event, or concept. Example: a specific student "Ram Bahadur" with ID 101.

Entity Set

Entity Set: An entity set is a collection of entities of the same type that share the same properties or attributes. Example: The set of all students enrolled in a university.

In an E-R diagram, an entity set is represented by a rectangle.

Types of Entities

Type	Symbol in ERD	Description
Strong Entity	Rectangle (single border)	Has a primary key; exists independently. Example: Student, Employee, Department.
Weak Entity	Rectangle (double border)	Cannot be uniquely identified by its own attributes alone; depends on a strong (owner/identifying) entity. Example: Dependent (of Employee), Order Item (of Order).

2.4 Relationship and Relationship Sets

Relationship

Relationship: A relationship is an association among two or more entities. Example: Student "enrolls in" Course — "enrolls in" is the relationship.

Relationship Set

Relationship Set: A relationship set is a set of relationships of the same type. Example: All (student, course) enrollment pairs form the "Enrolls" relationship set.

In an E-R diagram, a relationship set is represented by a diamond shape.

Degree of a Relationship

Degree	Description
Unary (Degree 1)	A relationship between instances of the same entity set. Example: Employee "manages" Employee.
Binary (Degree 2)	A relationship between two entity sets. Most common. Example: Student "enrolls in" Course.
Ternary (Degree 3)	A relationship involving three entity sets. Example: Supplier "supplies" Part to Project.

Mapping Cardinalities (Cardinality Constraints)

Mapping cardinality expresses the number of entities to which another entity can be associated via a relationship set.

Cardinality	Meaning	Example
One-to-One (1:1)	One entity in A is associated with at most one entity in B, and vice versa.	One person has one passport; one passport belongs to one person.
One-to-Many (1:N)	One entity in A is associated with many entities in B; each entity in B is associated with at most one in A.	One department has many employees; each employee belongs to one department.
Many-to-One (N:1)	Many entities in A are associated with one entity in B.	Many employees work for one company.
Many-to-Many (M:N)	Many entities in A are associated with many entities in B.	Students enroll in many courses; each course has many students.

Participation Constraints

Type	Description	E-R Notation
Total Participation	Every entity in the entity set must participate in at least one relationship instance. (Mandatory)	Double line in E-R diagram
Partial Participation	Some entities may not participate in any relationship instance. (Optional)	Single line in E-R diagram

Example: Every loan must be associated with at least one customer (total participation of Loan in Borrower relationship). A customer may or may not have a loan (partial participation of Customer).

2.5 Strong and Weak Entity Sets

Exam Tip: "What is a weak entity set and identifying relationship? Explain with example." [4+4] and "Explain strong and weak entity sets along with example." [8+4] — very frequently asked.

Strong Entity Set

Strong Entity Set: A strong entity set has sufficient attributes to form a primary key — it can be uniquely identified by its own attributes alone.

- Represented by a single rectangle in E-R diagram
- Has a primary key (one or more attributes)
- Exists independently — does not depend on any other entity
- Example: Student (sid, name, age), Employee (empid, name, salary), Department (dept_id, dept_name)

Weak Entity Set

Weak Entity Set: A weak entity set does not have sufficient attributes to form a primary key. It is identified only by being related to a strong (owner) entity and using a partial key (discriminator).

- Represented by a double rectangle in E-R diagram
- Has a discriminator/partial key (underlined with dashed line) rather than a primary key
- Its full key = primary key of owner entity + partial key
- Connected to its owner via an identifying relationship (double diamond)
- Example: Dependent (of Employee) — identified by (empid + dependent_name); Order_Item (of Order) — identified by (order_id + item_no)

	Strong Entity	Weak Entity
Key	Has a primary key	Has only a partial key (discriminator)
Existence	Exists independently	Depends on an owner/identifying entity
E-R Symbol	Single rectangle	Double rectangle
Relationship	Normal relationship (single diamond)	Identifying relationship (double diamond)
Example	Employee, Student, Product	Dependent of Employee, Order Item of Order

Identifying Relationship

Identifying Relationship: The relationship that associates a weak entity set with its owner (strong) entity set. Represented by a double diamond in E-R diagram.

Example: "Has-Dependent" is the identifying relationship between Employee (strong) and Dependent (weak).

Discriminator (Partial Key)

Discriminator / Partial Key: The attribute(s) that distinguish among all the entities in the weak entity set that depend on one particular strong entity. It is underlined with a dashed line in the E-R diagram.

Example: For the weak entity Dependent depending on Employee, the discriminator might be "dependent_name". Two dependents of the same employee cannot have the same name, but dependents of different employees can.

2.6 Attributes and Keys

Exam Tip: "Explain the different types of attributes. What is a discriminator key?" [2+2] — frequently asked. Also: "Explain different types of keys in a relational database along with example." [4]

Types of Attributes

Attribute Type	Description	Example
Simple (Atomic) Attribute	Cannot be divided into smaller subparts. The most basic type.	Age, Gender, Roll Number
Composite Attribute	Can be divided into smaller subparts (sub-attributes). The parts have independent meaning.	Name = (FirstName, MiddleName, LastName); Address = (Street, City, ZIP)

Single-Valued Attribute	Holds only one value for a given entity.	Age, Date of Birth, Student ID
Multi-Valued Attribute	Can hold multiple values for a given entity. Represented by double ellipse in ERD.	Phone Numbers (a person may have many), Email Addresses, Skills
Derived Attribute	Value can be computed/derived from other attributes. Represented by dashed ellipse in ERD.	Age (derived from Date of Birth); Duration (derived from Start and End Date)
Stored Attribute	Value is physically stored in the database (opposite of derived).	Date of Birth (used to derive Age)
Null Attribute	An attribute that may have a null value (no value, unknown, or not applicable).	Middle Name (some people may not have one)
Complex Attribute	A combination of composite and multi-valued attributes.	Address_History (multiple addresses, each composite)
Key Attribute	An attribute whose value uniquely identifies each entity in the entity set. Underlined in ERD.	Student ID, Employee ID, Roll Number
Descriptive Attribute	An attribute that belongs to a relationship set rather than an entity set.	Grade (in Student-Course Enrolls relationship), Date (in Person-Accident Participated)

Types of Keys

Key Type	Definition	Example
Super Key	A set of one or more attributes that, taken collectively, allows identification of a unique entity. May contain redundant attributes.	(sid), (sid, name), (sid, name, age) — all are super keys of Student
Candidate Key	A minimal super key — no proper subset of a candidate key is also a super key. A relation may have multiple candidate keys.	(sid) and (email) could both be candidate keys of Student
Primary Key	The candidate key selected by the DBA to uniquely identify entities in the entity set. Only one per entity set.	sid is chosen as primary key of Student. Underlined in ERD.
Composite Key	A primary key consisting of more than one attribute.	(student_id, course_id) as primary key of Enrollment table
Foreign Key	An attribute in one table that references the primary key of another table. Enforces referential integrity.	dept_id in Employee table references dept_id in Department table
Alternate Key	A candidate key that was not selected as the primary key.	If sid is PK of Student, then email is an alternate key

Partial Key / Discriminator	An attribute of a weak entity set that distinguishes weak entities related to the same owner. Underlined with dashed line.	dependent_name in Dependent (depends on Employee)

2.7 E-R Diagram

Exam Tip: Drawing ER diagrams is worth 8 marks in TU exams. The same scenarios (Company, University, Hospital, Library, Airline, etc.) repeat across years. Master the notation and practice all scenarios below.

E-R Diagram Notation Summary

Symbol	Shape/Notation	Represents
Entity Set	Rectangle (single border)	A strong entity set (e.g., Student, Employee)
Weak Entity Set	Rectangle (double border)	A weak entity set (e.g., Dependent, Order Item)
Relationship Set	Diamond (single border)	A relationship set between entities
Identifying Relationship	Diamond (double border)	Relationship between weak entity and its owner
Attribute	Ellipse (single border)	A simple or single-valued attribute
Multi-Valued Attribute	Ellipse (double border)	An attribute with multiple values (e.g., phone numbers)
Derived Attribute	Dashed ellipse	An attribute derived from other attributes (e.g., age)
Key Attribute	Ellipse with underlined text	Primary key attribute of an entity set
Partial Key	Ellipse with dashed underlined text	Discriminator of a weak entity set
Total Participation	Double line (===)	Every entity participates in the relationship (mandatory)
Partial Participation	Single line (—)	Some entities may not participate (optional)
Cardinality (1:1, 1:N, M:N)	Labels on relationship lines	Type of mapping between entity sets

Steps to Draw an E-R Diagram

- Step 1: Identify all entities (nouns in the problem statement that are independent objects)
- Step 2: Identify attributes for each entity; mark the primary key (underline)
- Step 3: Identify relationships between entities (verbs/associations)
- Step 4: Determine cardinality (1:1, 1:N, M:N) for each relationship

- Step 5: Determine participation (total or partial) for each entity in each relationship
- Step 6: Identify any weak entities and their identifying relationships
- Step 7: Add descriptive attributes to relationships if needed

Common ER Diagram Scenarios from Past Papers

ERD TEMPLATE: Hospital Management System (2082 Kartik)
DEPARTMENT (dept_id [PK], dept_name)
DOCTOR (doctor_id [PK], name, specialization, experience) — belongs to one DEPARTMENT (N:1)
PATIENT (patient_id [PK], name, age, gender, medical_history) — treated by DOCTOR (M:N)
ROOM (room_no [PK], type: ICU/General/Private, cost_per_day) — patient admitted to ROOM (M:1)
NURSE (emp_id [PK], name, shift_timings) — assists DOCTOR and cares for PATIENT
MEDICINE (medicine_id [PK], name, expiry_date) — prescribed by DOCTOR to PATIENT (with quantity)
APPOINTMENT (patient_id + doctor_id + datetime as composite key, date, time)
Relationships: DEPARTMENT has DOCTOR (1:N), DOCTOR treats PATIENT (M:N), PATIENT admitted to ROOM (M:1), NURSE assists DOCTOR (M:N), DOCTOR prescribes MEDICINE to PATIENT (ternary or descriptive)
Note: *Total participation: Every appointment must involve a doctor and patient. Every patient admitted must have a room.*

ERD TEMPLATE: Company / Project Database (2081 Chaitra, 2079 Chaitra)
EMPLOYEE (empid [PK], name, address, salary) — long-term employees only, no temps
DEPARTMENT (dept_id [PK], dept_name) — headed by a manager (one EMPLOYEE manages one DEPARTMENT)
PROJECT (project_id [PK], project_name, budget, location)
SKILL (skill_id [PK], skill_name)
Relationships:
EMPLOYEE works_in DEPARTMENT (M:1 — many employees per dept)
EMPLOYEE manages DEPARTMENT (1:1 — one manager per dept)
EMPLOYEE works_on PROJECT (M:N with hours as descriptive attribute)
EMPLOYEE has SKILL (M:N)
PROJECT belongs_to DEPARTMENT (M:1)
EMPLOYEE supervises EMPLOYEE (unary/recursive — manager relationship)

Note: *This scenario repeats in nearly every exam year under different contexts (Company, IT firm, Engineering company).*

ERD TEMPLATE: Library Database (2081 Chaitra Q4)
BOOK (BID [PK], Title, Author, Genre, Publisher, Year)
MEMBER (MID [PK], Name, Email, Phone, Address)
LIBRARIAN (LibID [PK], Name, Email, Phone, Salary)
Relationships:
MEMBER issues BOOK via IssuedBook (LibID, MID, BID, IssueDate, ReturnDate) — M:N with descriptive attributes
LIBRARIAN manages BOOK via Manages (LibID, BID) — M:N
Note: *IssuedBook is an associative entity (relationship with its own attributes).*

ERD TEMPLATE: University / Registrar Database (2078 Chaitra)
DEPARTMENT (dept_id [PK], dept_name)
INSTRUCTOR (id [PK], name, dept_id, title) — belongs to one DEPARTMENT
COURSE (course_no [PK], title, credits, syllabus, prerequisites)
COURSE_OFFERING (offering_id [PK], course_no, year, semester, section_no, instructor_id, timing, classroom)
STUDENT (student_id [PK], name, program)
Relationships:
DEPARTMENT offers COURSE (1:N)
INSTRUCTOR teaches COURSE_OFFERING (1:N)
STUDENT enrolls_in COURSE_OFFERING (M:N, with grade as descriptive attribute)
INSTRUCTOR heads DEPARTMENT (1:1)
Note: *Grade is a descriptive attribute on the Enrollment relationship, not on Student or Course.*

ERD TEMPLATE: Online Auction System (2079 Jestha)
MEMBER (member_no [PK], email, name, password, home_address, phone) — can be buyer and/or seller
BUYER (member_no [FK], shipping_address) — subtype of Member
SELLER (member_no [FK], bank_account_no, routing_no) — subtype of Member
ITEM (item_no [PK], title, description, starting_bid_price, bidding_increment, start_date, end_date) — placed by SELLER
CATEGORY (category_id [PK], name, parent_category) — items categorized hierarchically

BID (member_no + item_no + bid_time as composite key, bid_price)
TRANSACTION (transaction_id [PK], buyer_id, seller_id, item_no, feedback_rating, comment)
Relationships: SELLER places ITEM (1:N), BUYER bids_on ITEM (M:N), ITEM categorized_as CATEGORY (M:N), MEMBER participates_in TRANSACTION (M:N)
Note: *Generalization: Member generalizes to Buyer and Seller (a member can be both).*

Generalization and Specialization in ER Diagrams

Generalization

Generalization is a bottom-up approach where two or more lower-level entity sets are combined to form a higher-level entity set based on their common features. Example: Car and Bus generalize to Vehicle.

Specialization

Specialization is a top-down approach where a higher-level entity set is divided into lower-level entity sets based on distinguishing characteristics. Example: Employee specializes into Permanent_Employee and Contract_Employee.

Aggregation

Aggregation is a concept where a relationship set is treated as an abstract entity. It is used when we need to express a relationship between a relationship and another entity. Example: A Manager "manages" the relationship between an Employee and a Project.

Concept	Direction	Purpose
Generalization	Bottom-up (specific to general)	Combine common attributes of multiple entities into one superclass
Specialization	Top-down (general to specific)	Divide a general entity into specialized subclasses
Aggregation	Groups a relationship + entities	Model relationships involving other relationships

2.8 Alternate Data Models

Exam Tip: "What are data models? Explain various types." and "Explain how network data model is different from relational model." [4] — asked in several years.

Hierarchical Data Model

In the hierarchical model, data is organized in a tree-like structure. Each record has one parent and can have multiple children. It represents one-to-many (1:N) relationships naturally.

- Developed by IBM in the 1960s (IMS — Information Management System)
- Root at the top; child nodes below; one parent per child
- Navigated using predefined parent-child paths
- Limitation: Cannot represent M:N relationships easily; data redundancy for shared children
- Example: File system directories (one folder can have many files/subfolders)

Network Data Model

The network model extends the hierarchical model by allowing records to have multiple parents, forming a graph (network) structure. It can represent M:N relationships.

- Developed by CODASYL (Conference on Data Systems Languages) in the 1970s
- Organizes data as a graph — any record can be related to any other
- Navigated using explicit pointers/links between records
- More flexible than hierarchical but very complex to program and maintain
- Example: A Course can have many Students AND a Student can take many Courses

Relational Data Model

The relational model organizes data into tables (relations) consisting of rows (tuples) and columns (attributes). It was proposed by E.F. Codd in 1970 and is the foundation for most modern DBMS (MySQL, PostgreSQL, Oracle, SQL Server).

- Data represented as tables (relations)
- Relationships expressed through primary and foreign keys
- Queries expressed in SQL (Structured Query Language)
- Supports M:N relationships through junction/associative tables
- Based on mathematical set theory and predicate logic

Object-Oriented Data Model

Integrates object-oriented programming concepts (objects, classes, inheritance, encapsulation, polymorphism) with database capabilities. Each entity is represented as an object.

- Supports complex data types (images, audio, video)
- Objects have both data (attributes) and behavior (methods)
- Inheritance allows sub-classes to inherit properties from parent classes
- Example: Poet++ (object-oriented DBMS)

Comparison of Data Models

Feature	Hierarchical	Network	Relational
Structure	Tree (1:N only)	Graph (M:N allowed)	Tables (rows & columns)
Relationships	Parent-child (single parent)	Multiple parents allowed	Foreign keys

Query Language	Procedural navigation	Procedural navigation	SQL (declarative)
Flexibility	Low	Medium	High
M:N Relationships	Not directly supported	Supported via pointers	Supported via junction table
Complexity	Simple but rigid	Complex	Moderate, well-understood
Example System	IBM IMS	IDMS	MySQL, Oracle, PostgreSQL

ER Diagram Drawing — Exam Answer Guide

Exam Tip: In TU exams, ER diagram questions carry 8 marks. Follow this structured approach to score full marks.

When writing an ER diagram answer in exams, include these sections:

1. Entities Identified:

List all entities with their key attributes. Underline the primary key.

2. Relationships:

List all relationships with cardinality (1:1, 1:N, M:N) and participation (total/partial).

3. Assumptions:

State any assumptions you make about the mapping constraints.

4. E-R Diagram:

Draw the diagram using proper notation (rectangles, diamonds, ellipses, lines).

ERD TEMPLATE: How to Handle Common Scenarios
Manager relationship: Use recursive/unary relationship on Employee entity
Supervisor relationship: Same — Employee supervises Employee (unary)
Subtype/Supertype: Use generalization (e.g., Employee -> Faculty, Staff, Student_Assistant)
Ternary relationship: When three entities participate in same relationship (e.g., Supplier-Part-Project)
Associative entity: When a relationship has its own attributes (turn diamond into rectangle too)
Total vs Partial participation: If problem says "must" or "required" -> total participation (double line)
If problem says "may" or "can" -> partial participation (single line)
Note: *Always justify your cardinality and participation choices with a brief note.*

Past Exam Questions — Chapter 2

Exam Tip: Chapter 2 carries 12 marks in TU exams — second highest. ER diagram (8 marks) + theory (4 marks) is the standard pattern.

ER Diagram Questions (8 marks each):

Q: Draw an ER-diagram for: Hospital Management System (multiple departments, doctors, patients, rooms, nurses, pharmacy, appointments).	**[8]**
Q: Design a relational database for a medium-sized Company (departments, employees, projects, skills, managers). Construct an ER diagram.	**[8]**
Q: Draw an ER-diagram for the mini-case: A publishing company produces books by authors. Editors edit book publications. Suppliers supply items. Shop owners buy books. Books are uniquely identified by BookID.	**[8]**
Q: Draw an ER-diagram for a Home Decor Company: salespersons, customers, orders, furniture items, suppliers, payments.	**[8]**
Q: Design an ER diagram for an ONLINE AUCTION database: members (buyers and sellers), items, categories, bids, transactions.	**[3+8]**
Q: Draw an ER-diagram for a database tracking employees, alumni, and students. Employees are faculty, staff, or student assistants. Graduate/undergraduate students. (Generalization/Specialization)	**[2+2+8]**
Q: Draw an ER diagram for an airline system tracking customers, reservations, flights, seat assignments, and future flight routing.	**[8]**
Q: Design an ER diagram for an NFL database: teams, players, coaches, matches.	**[8+4]**
Q: Draw an ER-diagram for a University Registrar: courses, offerings, students, instructors, enrollments.	**[8]**
Q: Draw an ER diagram for: A Bus Company with buses, routes, drivers, towns, and garages.	**[8+4]**

Q: Draw an ER diagram for an Engineering Company: employees with skills, machine-types, maintenance requirements.	**[8+4]**
Q: Draw an ER diagram for a pharmacy chain: patients, doctors, drugs, pharmacies, pharmaceutical companies, contracts, prescriptions.	**[8]**

Theory Questions:

Q: Explain the different types of attributes. What is a discriminator key?	**[2+2]**
Q: Explain different types of keys in a relational database along with examples.	**[4]**
Q: Explain the various elements of the relational data model along with an example.	**[3+1]**
Q: What are data models? Explain various types of data models.	**[1+3]**
Q: Define total and partial participation with example. Explain weak entity set with an example.	**[2+2]**
Q: What is a weak entity set and identifying relationship? Explain with example.	**[4]**
Q: Explain strong and weak entity sets along with example.	**[8+4]**
Q: Distinguish between degree and cardinality of a relationship in an ER diagram. Explain generalization in ER diagram with an example.	**[2+2]**
Q: Explain generalization and specialization in ER diagram along with an example.	**[4]**
Q: Define unary relationship along with example. How do you convert an ER relationship into a relation schema? Explain with examples of different cardinalities.	**[2+4]**
Q: Explain how network data model is different from relational data model.	**[4]**
Q: Define aggregation in ER design. Discuss with an example.	**[2+2]**

Quick Revision — Key Points Chapter 2

- Data Model = conceptual tools to describe data, relationships, semantics, and constraints.
- Three model levels: Conceptual (E-R), Logical (Relational), Physical (storage).
- E-R Model by Peter Chen (1976) — uses entities, attributes, relationships.
- Entity Set: rectangle | Relationship Set: diamond | Attribute: ellipse.
- Weak entity: double rectangle; Identifying relationship: double diamond; Partial key: dashed underline.
- Strong entity = has own PK; Weak entity = depends on owner + uses partial key/discriminator.
- Cardinality: 1:1, 1:N, N:1, M:N — how many entities associate.
- Participation: Total (double line, mandatory) vs Partial (single line, optional).
- Attribute types: Simple, Composite, Single-valued, Multi-valued (double ellipse), Derived (dashed ellipse).
- Key types: Super Key > Candidate Key > Primary Key | Also: Foreign Key, Alternate Key, Partial Key.
- Generalization (bottom-up, specific->general) vs Specialization (top-down, general->specific).
- Aggregation: treat a relationship as an entity to relate with another entity.
- Hierarchical model: tree, 1:N, IBM IMS. Network model: graph, M:N, CODASYL. Relational: tables, SQL.